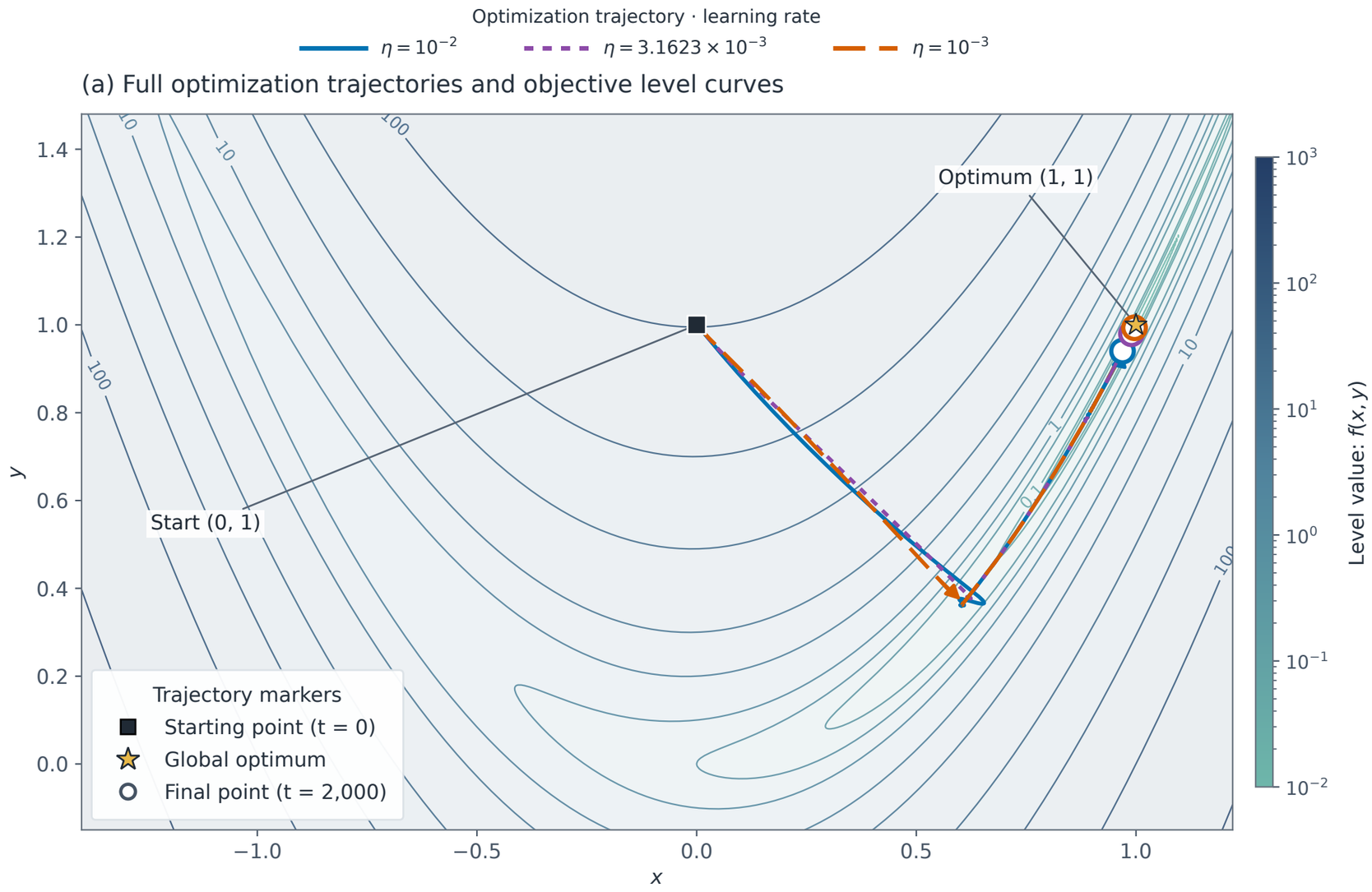
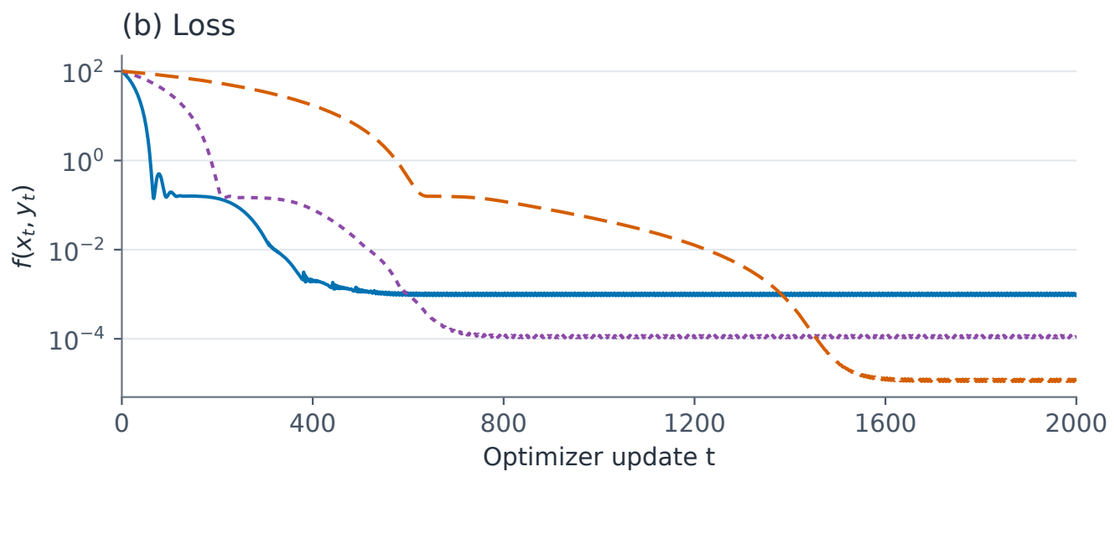
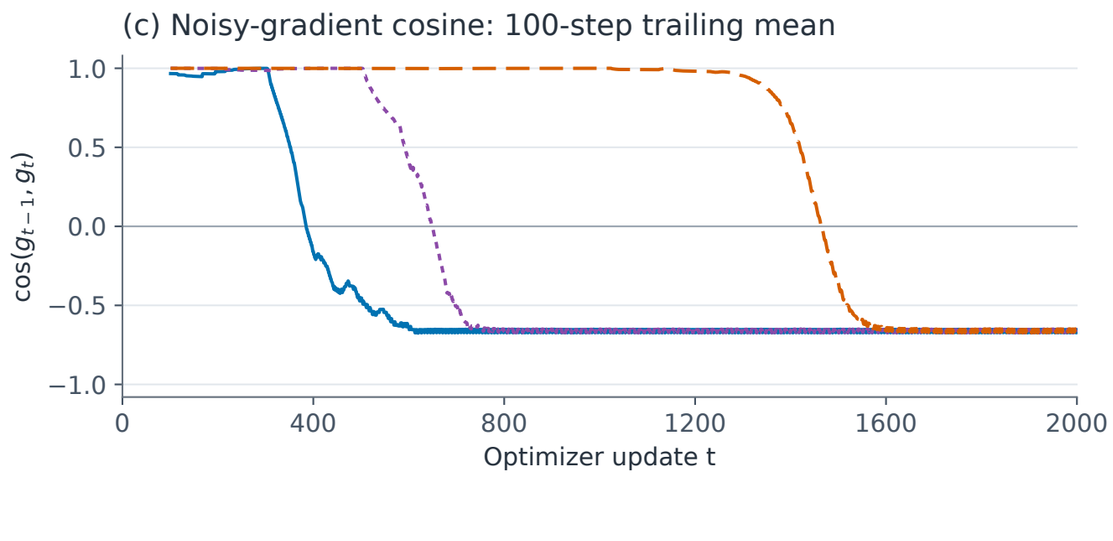
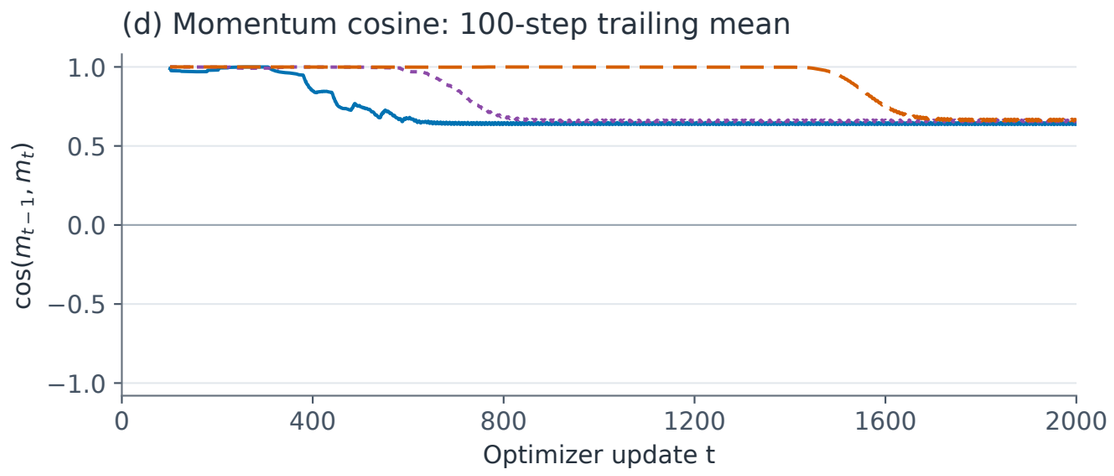
Rosenbrock · AdamW noisy gradients · σ = 0, weight decay 0.1

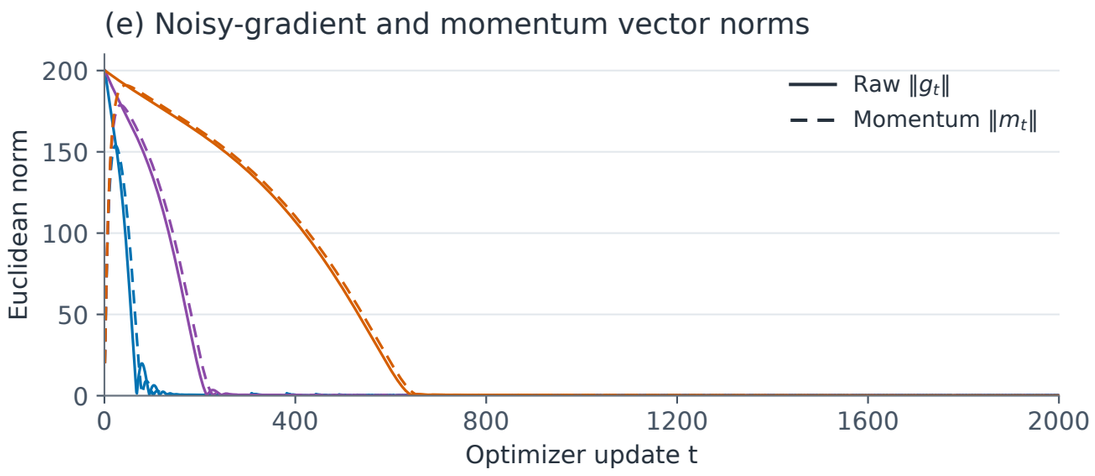
2,000 updates per learning rate · $\beta_1 = 0.9$, $\beta_2 = 0.95$ · CPU float64

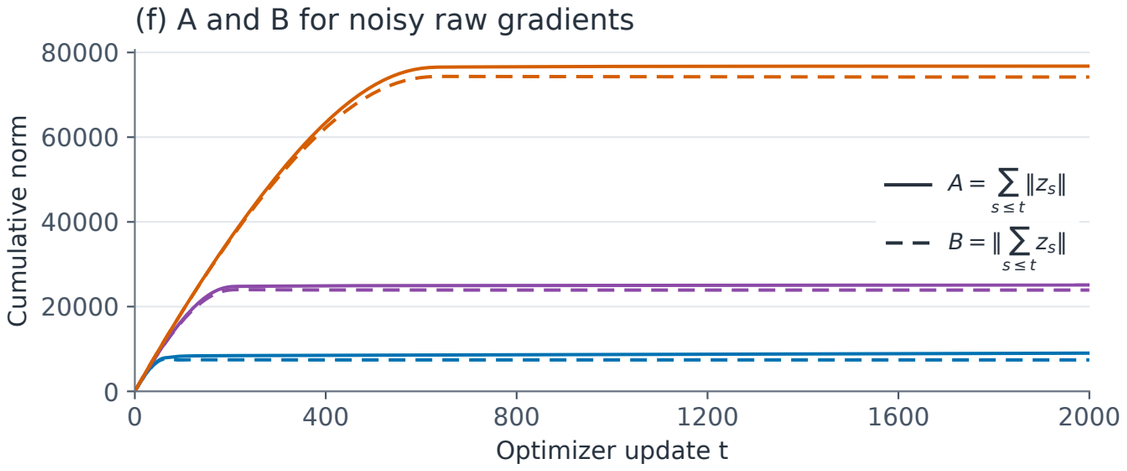


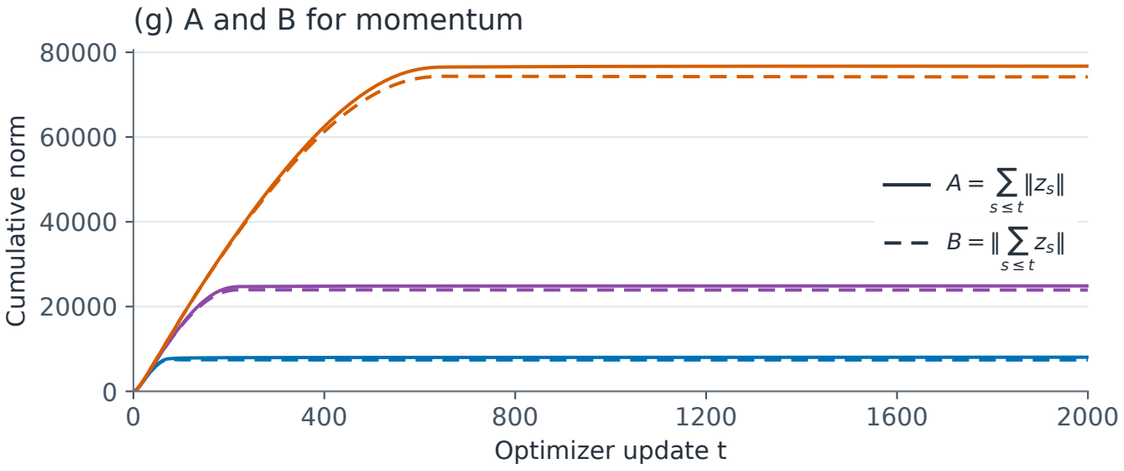


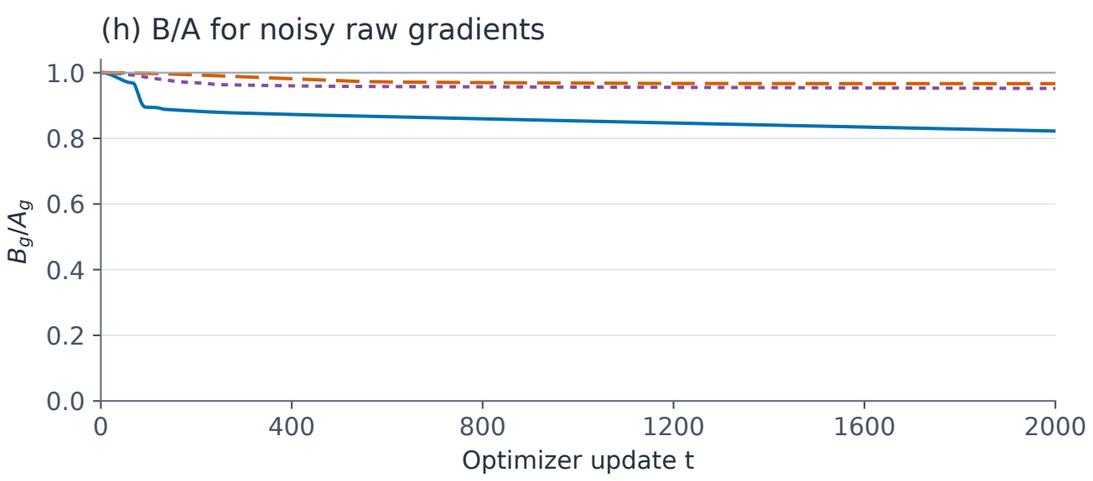


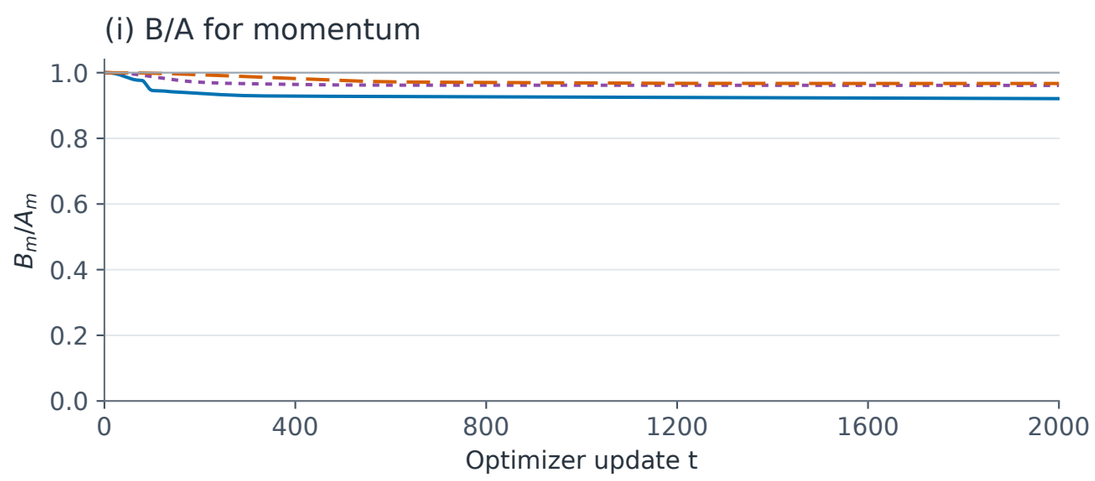












$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$. Panels (c,d): 100-step trailing means; complete finite windows only. Other series unsmoothed.

m_t is the stored, uncorrected first moment.

$g_t = \nabla f(x_{t-1}, y_{t-1}) + \xi_t$, $\xi_t \sim \mathcal{N}(0, \sigma^2 I_2)$; loss is exact. Reused deterministic controls. First cosine and B/A at A = 0 are undefined.